

MANOJ G K

Bangalore, India | manojgk0210@gmail.com | +91 8618682628 | [LinkedIn](#) |

PROFILE SUMMARY

Analytical and detail-oriented MCA student with hands-on experience in Python development, cloud deployment, and DevOps practices. Proficient in designing user-focused web applications, optimizing backend logic, and maintaining code quality. Adept at working with cloud platforms, automation tools, and collaborative workflows. Strong communicator with a passion for continuous learning, systems thinking, and delivering scalable solutions.

EDUCATION

Master of Computer Application (MCA)

NITTE Meenakshi Institute of Technology, Bangalore

November 2024 - Present

CGPA: 7.93

Bachelor of Computer Application (BCA)

Don Bosco Degree College, Chitradurga

June 2021 - August 2024

CGPA: 8.07

PROFESSIONAL EXPERIENCE

Building Scalable, Secure, and High-Performance Web Applications with AWS 3-Tier Architecture

- Designed and provisioned complete AWS infrastructure using Terraform, including S3, IAM roles, VPC, subnets, gateways, and routing.
- Created an S3 bucket to store application code and configured IAM roles with restricted read-only S3 access for EC2 instances.
- Built a custom VPC with multi-AZ architecture, including public subnets (web tier), private subnets (application tier), and private database subnets.
- Configured Internet Gateway, NAT Gateways, and Route Tables to enable secure access for public resources and outbound access for private instances.
- Deployed Application Load Balancers and Auto Scaling Groups for web and application tiers to ensure high availability and fault tolerance.
- Set up Amazon RDS (Multi-AZ) in private subnets, configured DB subnet groups, security groups, and validated connectivity using private EC2 instances and Session Manager.
- Created AMI images for web and application tiers, implemented scaling policies, and validated end-to-end application access via load balancer DNS.

Deployment of MERN Stack Using AWS, Terraform & Jenkins

- Designed and deployed a complete DevOps pipeline for a three-tier MERN application using AWS EKS with full automation, CI/CD, monitoring.
- Automated AWS infrastructure components (VPC, Private EKS Cluster, Worker Nodes, Jump Server) using Terraform, triggered via Jenkins pipelines.
- Built a CI pipeline integrating SonarQube, OWASP Dependency Check, file scanning, Docker build, and Trivy image scanning, pushing images to AWS ECR with automatic Kubernetes manifest updates.
- Implemented ArgoCD for GitOps-based CD, enabling continuous and automated deployment to the EKS cluster.
- Deployed MERN application on Kubernetes with MongoDB StatefulSets, Persistent Volumes, AWS Load Balancer Controller, and Ingress with a custom Route 53 domain.
- Established monitoring and observability using Prometheus and Grafana with pre-configured dashboards.
- Followed security best practices using IAM Roles for Service Accounts (IRSA) and a fully automated security scanning workflow.
- Stack: AWS (EKS, EC2, VPC, ECR, Route 53, IAM), Terraform, Kubernetes, Jenkins, ArgoCD, Docker, SonarQube, Trivy, Prometheus, Grafana, Helm, Bash, Node.js, React, MongoDB.

SKILLS

Programming & Tools:

- Python (Intermediate).
- Amazon Web Services (AWS)
- Git & GitHub – Version control, branching strategies, GitHub workflows.
- Terraform – Infrastructure provisioning and automation.
- Ansible – Configuration management and automation.
- Docker – Image creation, container lifecycle.
- Kubernetes – Pods, Services, Deployments, Cluster management.
- Linux – Basic command-line usage, file system navigation, user and permission management, package installation, and service handling.
- CI/CD pipelines – Jenkins, GitHub Actions.

Soft Skills:

- Problem-solving, teamwork, and communication.
- Time management and attention to detail.
- Critical Thinking.
- Self-Motivation & Curiosity.

CERTIFICATIONS

- AWS Certified Cloud Practitioner** – Jan 2025 to Mar 2025 | [\[Link\]](#)
- Cisco Certified: Networking Basics** – June 2025 | [\[Link\]](#)